# AI SKIN INTELLIGENCE AND PERSONALIZED SKINCARE PLANNER

Infosys Virtual Internship Report

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Technology stack: FastAPI · SQLAlchemy · SQLite · React · NumPy · Pillow · scikit-learn · XGBoost

# ABSTRACT

Choosing a skincare routine is difficult because skin condition depends on many interacting factors, including skin type, age, sleep, hydration, sun exposure and stress, and most people select products through advertising or trends without knowing whether they suit their skin or conflict with their allergies. Clinical advice is valuable but episodic, and individuals seldom keep a continuous record of how their skin changes. To address this issue, an AI Skin Intelligence and Personalized Skincare Planner has been developed as a full-stack web application.

The system captures a skin profile and daily lifestyle logs, optionally analyses single or three-angle facial images using NumPy and Pillow to measure smoothness, pores, oiliness, redness and fine lines, and produces an explainable skin health score from a five-factor weighted model together with ranked skin concerns and risk factors. A routine generator composes personalized morning, evening and weekly routines, a TF-IDF and cosine-similarity recommender ranks catalogue products, and an ingredient check flags allergens. Progress tracking, a skin-cycling and turnover view and a reminder centre support continued use, and a dermatologist portal allows a clinician to review patient data and issue a prescription.

The back end uses FastAPI, SQLAlchemy and SQLite with JWT authentication and PBKDF2 password hashing, and the front end is a React single-page dashboard. Functional testing of the REST API confirmed correct authentication, workflow and scoring behaviour, and offline XGBoost experiments on synthetic data were also reported. The system provides accessible, transparent and personalized skincare guidance.

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# CHAPTER 1 INTRODUCTION

Skincare decisions are made every day by millions of people, yet most are made without a structured understanding of the individual's own skin. Skin type, age, sensitivities, sleep, hydration, sun exposure and stress all influence how skin behaves, but these factors are rarely examined together. Product choices are therefore driven by advertising, trends or casual advice, and users find it difficult to know whether a routine is working or whether a product is safe for their allergies.

The AI Skin Intelligence and Personalized Skincare Planner collects a skin profile and daily lifestyle information, optionally analyses facial photographs using image-processing algorithms, and combines these inputs into a skin health assessment, a personalized morning, evening and weekly routine, ingredient-aware product recommendations, progress analytics and reminders. A dermatologist portal lets a clinical user review patient data and issue a prescribed routine. Every score and recommendation can be traced back to the inputs that produced it.

**Personalized skincare analytics** applies data-driven methods to health, lifestyle and appearance data so that care suits an individual rather than an average consumer. Its characteristics are: individualized assessment from profile and lifestyle data; explainable scoring in which each contributing factor is visible; objective measurement of skin-surface features from photographs; safety awareness through allergy checks; continuous tracking of progress and adherence; and timely reminders that support habit formation.

## 1.1 Problem Statement and Objectives

Individuals who wish to improve their skin often do not know how sleep, hydration, sun exposure and stress influence their concerns. Product catalogues are large and ingredient lists technical, so it is hard to match products to a skin type and avoid allergens. Routines are rarely tracked, and dermatological advice is available only in occasional consultations without a continuous record. A single accessible system is needed that gathers profile and lifestyle data, measures visible skin features, converts them into an understandable assessment and routine, recommends safe products, tracks progress, and lets a dermatologist review the same data. The objectives are:

- Secure registration and login with token-based authentication and four roles (user, skincare consultant, dermatologist, administrator).
- Capture of a structured skin profile (type, age group, concerns, allergies, sensitivities) and a daily lifestyle log.
- An explainable rule-based engine producing a skin health score, ranked concerns and risk factors.
- Single and three-angle facial image analysis giving texture, pore, oiliness, redness and fine-line metrics with visual overlays.
- Personalized morning, evening and weekly routines that respect concerns, sensitivities and scan findings.
- TF-IDF product recommendation with allergy-aware ingredient flagging.
- Progress, notification, replenishment and skin-cycling support through an interactive dashboard, plus a dermatologist portal for issuing prescriptions.

## 1.2 Scope of the Project

The system is a web application with a Python FastAPI back end, a SQLite database and a React front end. It processes only data entered by the user and images uploaded or captured through the browser. The assessment and routine logic is rule based, the catalogue is a curated set of 21 products, and the application is a skincare planning and awareness aid. **It is not a medical diagnostic device** and does not replace professional dermatological advice. Chapter 2 surveys the underlying techniques, Chapter 3 presents the design-thinking exercise, Chapter 4 the methodology, Chapter 5 the implementation, testing and results, and Chapter 6 the conclusion and future work.

# CHAPTER 2 LITERATURE SURVEY

The platform brings together machine learning, information retrieval, dermatology, web security and software architecture. The review concentrates on the foundational works behind the techniques that the source code actually relies on.

- **XGBoost [1]** – Chen and Guestrin introduce a scalable gradient-boosted tree system with regularized objectives, well suited to mixed numeric and categorical tabular data. The offline experiments train an XGBoost regressor (health score) and a multi-label classifier (four concerns).
- **Scikit-learn [2]** – Pedregosa et al. describe a library with consistent interfaces for preprocessing, pipelines and evaluation. The project uses ColumnTransformer, StandardScaler, OneHotEncoder, Pipeline, MultiOutputClassifier, TfidfVectorizer and cosine similarity.
- **TF-IDF [3]** – Salton and Buckley establish term frequency with inverse document frequency as an effective weighting scheme. Each product is treated as a short document and the user's skin type and concerns as a query, ranked by cosine similarity.
- **Fitzpatrick scale [4]** – a classification of sun-reactive skin types that motivates the sun-protection emphasis in the UV risk rules and sunscreen guidance.
- **REST [5]** – Fielding defines resource-oriented, stateless interfaces; the FastAPI back end exposes JSON endpoints in this style.
- **JWT [6]** – a compact signed token standard; an HS256 token carrying the user's e-mail as subject is issued at login and verified on every protected request.
- **PBKDF2 [7]** – password-based key derivation; passwords are stored using PBKDF2-HMAC-SHA-256 with a random salt, 100,000 iterations and constant-time comparison.
- **NumPy and pandas [8], [9]** – NumPy underpins the pixel-level texture computations; pandas generates and reads the offline experiment datasets.
- **SQLAlchemy [10]** – an object-relational mapper separating the domain model from the database engine; twelve ORM models run on SQLite or any other database via the connection URL.

It is observed that explainable rule-based scoring, content-based retrieval and lightweight image analysis can be combined without heavy deep-learning infrastructure, while REST, JWT and PBKDF2 provide a secure and maintainable service layer.

# CHAPTER 3 DESIGN THINKING

Empathy mapping was carried out for the two primary user groups before any interface or workflow was designed. Each map records what the user says, thinks, does and feels, so that the system is technically correct and genuinely user-centred.

| Quadrant | Persona 1 – Skincare User | Persona 2 – Dermatologist / Consultant |
|---|---|---|
| **Says** | "I am not sure what my skin type really is." "Which products suit my concerns and allergies?" "Is my routine actually working?" | "I need the patient's skin data before the consultation." "Which patients show concerning scan results?" "I want to prescribe a routine that suits the patient." |
| **Thinks** | Do sleep, water and sun habits affect my skin? Is this product safe for my allergies? Can I follow a routine every day? | Is the patient's lifestyle affecting the condition? Which catalogue products are appropriate? Will the patient follow the plan? |
| **Does** | Fills in a profile; logs sleep, water, UV and stress; scans the face; checks ingredients; tracks progress. | Reviews the patient panel; browses the clinical and trendy catalogue; issues a prescription with diagnosis, notes and routines. |
| **Feels** | Confused by the number of products and claims; worried about reactions; motivated when progress is visible. | Limited by episodic visits; confident when advice is backed by measurements; concerned about allergies. |

**Pain points and gains.** The user suffers from product confusion, allergy worries, no record of daily habits and forgotten steps; the system answers with an explainable score, a personalized routine, allergy-aware recommendations, a progress trend and reminders. The clinician lacks continuous data between visits; the system offers a patient panel with the latest assessment and scan, a product catalogue, a prescription form, and a prescription that appears directly in the patient's account.

**Design implications.** A guided wizard removes the need for technical terms, the lifestyle log turns habits into measurable inputs, the engine explains every score, the ingredient check addresses allergy fears, and the tracker and reminders keep the routine consistent. The dermatologist portal responds to the need for continuous, measured patient data.

# CHAPTER 4 PROPOSED METHODOLOGY

## 4.1 Existing and Proposed System

**Existing system.** Skincare guidance today is fragmented: retail sites recommend by popularity or advertising, articles prescribe the same routine to everyone, and many applications offer only a fixed checklist or a product scanner. They rarely combine profile, habits and measured skin features, give no allergy warning, and keep no consolidated record. Clinical consultation is authoritative but episodic, with no continuous record of sleep, hydration, sun protection or earlier scans, and reminders are handled informally.

**Proposed system.** An integrated, explainable, secure web platform developed in stages: (1) a secure account system and guided profile wizard; (2) a daily lifestyle logger for sleep, water, UV, pollution, stress and sunscreen use; (3) an optional computer-vision module measuring texture, pores, oiliness, redness and fine lines from one or three images. These feed a rule-based Assessment Engine that produces ranked concerns, risk factors and a five-factor health score, which drives the Routine Generator. A TF-IDF recommender ranks products, an ingredient analyzer flags allergens, and a progress tracker, notification centre and skin-cycling view support continued use. A dermatologist portal issues prescriptions that appear in the patient's dashboard.

## 4.2 System Design

**Use case diagram (Fig. 4.1).** The user (patient) registers, creates a skin profile, logs lifestyle data, scans skin texture, runs assessments, generates routines, views recommendations, checks ingredient safety, tracks progress, manages notifications and views the skin cycle analysis. The dermatologist or consultant views the patient panel, browses the catalogue and issues prescriptions.

**Entity relationship diagram (Fig. 4.2).** Data is stored in twelve SQLAlchemy tables with the users table as root; every user-specific table refers to it, so deleting a user removes dependent records. Variable-length data such as concerns, allergies, routine steps and score breakdowns is held in JSON columns. A user has at most one skin profile and one notification-settings record, and any number of lifestyle logs, assessments, routines, texture analyses, progress logs, notifications and replenishment records. The prescriptions table links to users twice (dermatologist_id and patient_id), and the products table is a standalone catalogue.

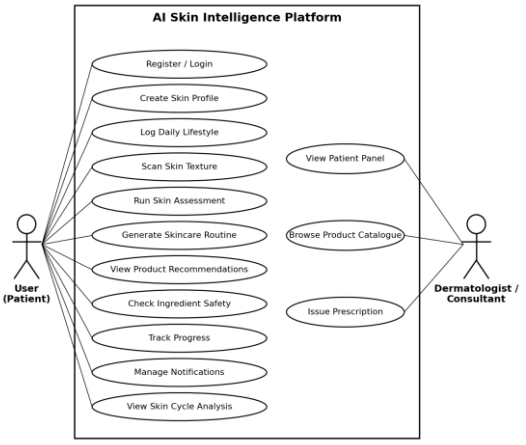



Fig. 4.1 Use Case Diagram

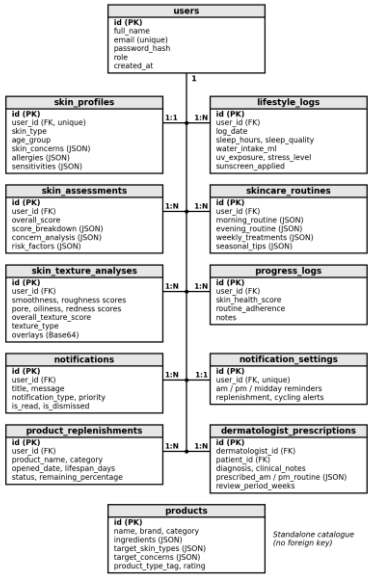



Fig. 4.2 Entity Relationship Diagram

## 4.3 System Architecture

The system follows a layered client–server architecture (Fig. 4.3). The presentation layer is a React single-page application built with Vite and styled with Tailwind CSS, which talks to the back end only through REST endpoints with the JWT as a bearer token. The API layer is a FastAPI application registering ten routers, validating requests with Pydantic schemas and enabling cross-origin requests. Business logic lives in service and machine-learning modules, and all persistence goes through SQLAlchemy to SQLite. The

workflow (Fig. 4.4) runs from registration and profile, through lifestyle log and optional scan, to assessment, routine, recommendation, tracking and clinical review.

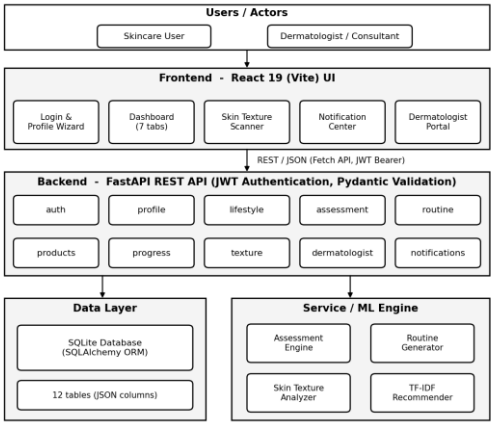



**Fig. 4.3 System Architecture**

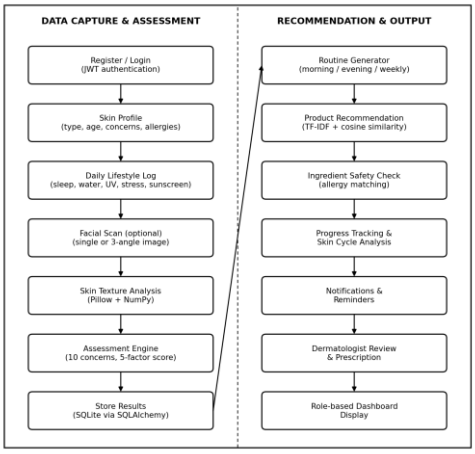



**Fig. 4.4 Workflow of the System**

Table 4.1 summarizes the REST interface. Every endpoint except registration, login, the mock sign-in, the product catalogue listing and the health checks requires a valid bearer token.

**Table 4.1 REST API Endpoints by Router**

| Router (prefix) | Endpoints | Purpose |
|---|---|---|
| Auth /api/auth | POST /register, /login, /oauth-mock; GET /me | Account creation, token issue, simulated social sign-in |
| Profile /api/profile | GET, POST | Read and create/update skin profile |
| Lifestyle /api/lifestyle | GET /today, POST, GET /history, /stats | Daily log, history, aggregated statistics |
| Assessment /api/assessment | POST /run; GET /latest, /history | Run and retrieve assessments |
| Routine /api/routine | POST /generate; GET /latest, /{type} | Generate and retrieve routines |
| Products /api/products | POST /; GET /, /recommendations, /{id}/analysis | Catalogue, recommendations, ingredient safety |
| Progress /api/progress | POST /; GET /, /analytics | Progress logging and trend analytics |
| Texture /api/texture | POST /analyze, /analyze-multi, /sync-profile; GET /history | Single and three-angle analysis, scan history |
| Dermatologist /api/dermatologist | GET /patients, /catalog, /my-prescription; POST /prescribe | Patient panel, catalogue, prescriptions |
| Notifications /api/notifications | GET; POST read, read-all, dismiss; settings; replenishments | Alerts, reminders, replenishment tracking |
| Utility | GET /, /health, /database-test | Service information and health checks |

## 4.4 Modules

The platform has nine modules: authentication and roles; skin profile and lifestyle tracking; skin assessment engine; skin texture vision analysis; routine generator; product intelligence; progress and skin cycle analysis; notifications and reminders; and the dermatologist portal.

### *1. Authentication and Role Management*

Registration takes full name, e-mail, password (minimum six characters) and role. The password is hashed with PBKDF2-HMAC-SHA-256 using a random 16-byte salt and 100,000 iterations, stored as scheme$iterations$salt$hash, and verified by recomputation with a constant-time comparison. Login returns an HS256 JWT whose subject is the e-mail and whose lifetime defaults to 1,440 minutes; a dependency decodes it on each protected request. Roles are USER, SKINCARE_CONSULTANT, DERMATOLOGIST and ADMIN. A simulated social sign-in creates or retrieves an account for demonstration without an external identity provider.

### *2. Skin Profile and Lifestyle Tracking*

The profile stores skin type (Dry, Oily, Combination, Normal, Sensitive), age group (Under 18 to 55+), concerns, allergies and sensitivities, created through a three-step wizard. The lifestyle logger records one entry per user per day (unique constraint on user and date): sleep hours (0–24) and quality, water intake in ml, UV and pollution exposure, stress level and sunscreen use. A statistics endpoint aggregates the latest 30 logs into average sleep and water, sunscreen compliance, and the most frequent UV and stress levels, which feed the assessment engine.

### *3. Skin Assessment Engine*

A deterministic rule-based module producing ranked concerns, risk factors and a five-factor health score. Each of ten concerns (Acne, Hyperpigmentation, Dark Spots, Dry Skin, Oily Skin, Sensitive Skin, Wrinkles, Fine Lines, Redness, Uneven Skin Tone) scores 0–10 (capped) by adding contributions from five evidence sources (Table 4.2). Severity is Critical (7–10), Moderate (4–6) or Low (1–3). Every contribution is stored as a readable modifier so the user can see why a concern scored as it did.

**Table 4.2 Evidence Sources Used in Concern Scoring**

| Source | Rule (as implemented) |
|---|---|
| Profile | Concern selected by the user: +4 |
| Skin type | Oily: Acne +2, Oily Skin +3, Hyperpigmentation +1, Uneven Tone +1. Dry: Dry Skin +3, Wrinkles +2, Fine Lines +2, Uneven Tone +1. Combination: Acne, Oily, Dry +1 each. Sensitive: Sensitive Skin +3, Redness +2, Uneven Tone +1 |
| Age group | Under 18 / 18–24: Acne +2. 25–34: Fine Lines +1. 35–44: Fine Lines, Wrinkles, Hyperpigmentation, Uneven Tone +1. 45–54 / 55+: Wrinkles, Fine Lines +2; Hyperpigmentation, Dark Spots +1 |
| Lifestyle | High/Moderate UV with sunscreen rate below 60%: pigmentation +2, Wrinkles/Fine Lines +1. High stress: Acne +2, Redness +1 (Medium: Acne +1). Sleep below 6 h: Dark Spots, Fine Lines +2, Uneven Tone +1. Water below 1,500 ml: Dry Skin +2, Fine Lines, Uneven Tone +1 |
| Texture scan | Primary concern match +3. Roughness >35: Uneven Tone, Dry Skin +2/+3. Redness >30: Redness, Sensitive +2/+3. Pore visibility >35: Acne, Oily +2/+3. Oiliness >40: Oily +2/+3. Fine lines >30: Fine Lines, Wrinkles +2/+3 |

The overall score is the weighted sum of five factors, each 0–100 (Table 4.3). With a scan, Skin Condition uses the overall texture score; otherwise it is 90 − 6 per concern − 4 per sensitivity, bounded 25–100. The score is summarized as excellent (≥85), good (70–84), fair (55–69) or needing dedicated care (<55). Risk factors (each with severity, description and action) include UV-induced damage, stress-related flares, skin repair deficit (sleep <6 h), dehydration, high sensitivity reactivity (≥3 sensitivities), insufficient data (<3 logs), and scan-based risks such as compromised surface texture or elevated erythema.

**Table 4.3 Five-Factor Skin Health Score Model**

| Factor | Weight | How the factor score is derived |
|---|---|---|
| Skin Condition Assessment | 35% | Texture scan score, or 90 − 6 × concerns − 4 × sensitivities (25–100) |
| Lifestyle Habits | 20% | Starts at 80; UV High −20, Moderate −10; stress High −15, Medium −5; bounded 20–100 |
| Sleep Quality | 15% | 7–9 h: 100; 6–7 h: 80; 5–6 h: 55; below 5 h: 30 |
| Routine Consistency | 20% | Sunscreen compliance rate (0–100) |
| Hydration Level | 10% | ≥2,500 ml: 100; 2,000: 85; 1,500: 65; 1,000: 45; below 1,000: 25 |

### *4. Skin Texture Vision Analysis*

Facial images are analysed using NumPy and Pillow only, with no pre-trained deep-learning model (pipeline in Fig. 4.5). The algorithm is:

1. **Input:** image bytes from a file upload or a Base64 string from the camera.
2. **Pre-process:** resize to 512 × 512 (LANCZOS); convert to RGB and grayscale arrays.
3. **Skin mask:** pixels with R>45, G>30, B>15, R>G, G>0.7B, R−G>5; if under 15% qualify, use the central 70% of the image.

4. **Roughness/smoothness:** central-difference gradient magnitude (mean, standard deviation, 90th percentile) gives roughness (10–95); smoothness = 100 − roughness.
5. **Pores:** subtract a Gaussian-blurred copy (radius 2.5) to get a high-pass response; pixels above the 82nd percentile inside the mask are pores, giving pore visibility and density.
6. **Oiliness/shine:** in HSV, the fraction of skin pixels with V>0.78 and S<0.38.
7. **Redness:** R − (G+B)/2 inside the mask (mean and 95th percentile).
8. **Fine lines:** variance of the horizontal gradient combined with roughness.
9. **Overall texture score:** 0.40×smoothness + 0.25×(100 − pore visibility) + 0.15×oiliness balance + 0.20×(100 − redness), bounded 10–99.
10. **Classify and visualize:** assign texture type and primary concern by thresholds; generate roughness, pore and redness overlays, zone-wise observations, recommended ingredients and action steps.

In three-angle mode the front, left and right images are analysed separately: smoothness is weighted 40/30/30, pore visibility and oiliness 50/25/25, redness and fine lines about equally, and the overall score is the mean of the three. The left–right smoothness difference is reported as a bilateral symmetry observation. Results, including Base64 overlays, are stored and used by the assessment engine and routine generator.

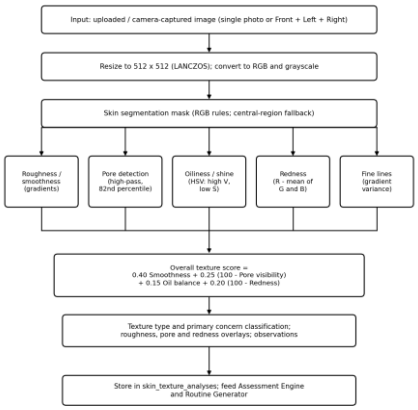



**Fig. 4.5 Skin Texture Analysis Pipeline**

### *5. Personalized Routine Generator*

The generator takes the profile, ranked concerns, allergies, sensitivities and latest texture data. The user is treated as sensitive if the skin type is Sensitive, Sensitive Skin is a prioritized concern, two or more sensitivities exist, or scan redness exceeds 40. It composes a morning routine (cleanser, optional toner, optional serum, moisturizer, sunscreen), an evening routine, weekly treatments and seasonal tips chosen by month. Each step names a product, key ingredients, reason, application method and expected benefit. A knowledge base of sixteen actives (salicylic acid, niacinamide, vitamin C, hyaluronic acid, retinol, azelaic acid and others) records benefits, contraindications and AM/PM suitability. Routines are stored so the latest can be retrieved.

### *6. Product Intelligence*

The catalogue holds 21 products (11 Dermatology Clinical, 10 Market Trendy) across serum, cleanser, moisturizer, sunscreen, exfoliant and toner, each with ingredients, target skin types and concerns, price band, rating, usage and key benefit. The TF-IDF recommendation algorithm is:

11. **Training:** build a document per active product from category, ingredients, target skin types and concerns (concerns repeated twice); fit a TF-IDF vectorizer with English stop words and save it with joblib.
12. **Query:** build the user document from skin type and concerns (concerns repeated three times).
13. **Vectorize and compare:** transform query and products with the saved vectorizer; compute cosine similarity.
14. **Rank:** return the top five with similarity above 0.01 (else the first five products).
15. **Safety check:** compare every ingredient with the user's allergies by case-insensitive substring match; return safe ingredients, flagged ingredients and an is-safe-to-use indicator.

### *7. Progress Tracking and Skin Cycle Analysis*

Progress entries store a skin health score, routine adherence and notes. The analytics endpoint returns the log count, average score, a trend (improving, declining or neutral by comparing first and last scores) and chart data. The Skin Cycle Analysis view has three sub-views: a skin journey dashboard; a four-night skin-cycling protocol (exfoliation night, cellular renewal night and two lipid-replenishment recovery nights) with a seven-day AM/PM plan; and a 28-day epidermal turnover tracker in four phases.

### *8. Notification and Reminder*

A synchronization routine creates routine reminders (morning), cycling directives (night) and replenishment alerts. Settings store on/off state and times for morning, evening and midday SPF reminders plus replenishment and cycling alerts. A replenishment record holds product, bottle size, opening date, lifespan and daily usage; the remaining percentage gives a status of GOOD, LOW or EMPTY. Notifications can be read, read-all or dismissed and may carry an action link to a dashboard tab.

### *9. Dermatologist Clinical Portal*

Dermatologist and consultant users see a clinical portal instead of the patient dashboard. It lists patients with profile, latest assessment and texture scan, shows the catalogue, and lets the clinician issue a prescription with diagnosis, clinical notes, clinical focus, review period, prescribed morning and evening routines and optional treatments. The patient retrieves the latest prescription through a dedicated endpoint.

## 4.5 System Requirements

The project needs a Python runtime for the back end, Node.js with npm to build the front end (static files by Vite), and a modern browser; it was developed on Windows with a Python virtual environment. A webcam is optional and used only for live capture; images can be uploaded instead. Python implements every back-end component, and its scientific libraries allow compact pixel-level computation. Table 4.4 lists the libraries.

**Table 4.4 Back-End and Front-End Libraries and Their Roles**

| Library | Role in the project |
|---|---|
| FastAPI, Uvicorn, Pydantic | REST API framework and ASGI server with interactive docs; request/response schemas and validation |
| SQLAlchemy | ORM models, sessions and table creation; SQLite delivered, others via connection URL |
| python-jose; hashlib, secrets | HS256 JWT creation/verification; PBKDF2-HMAC-SHA-256 hashing and constant-time comparison |
| python-dotenv, python-multipart | Loading DB URL, secret key, token lifetime from .env; image uploads |
| Pillow, NumPy | Image decoding, resizing, filtering and all pixel-level skin measurements |
| scikit-learn, XGBoost | TF-IDF, cosine similarity, pipelines, metrics; offline boosted regressor and classifier |
| pandas, joblib | Dataset generation/loading; saving and loading trained models |
| React 19, Vite, Tailwind CSS 4 | Component-based SPA (each tab a component); dev server and build; utility-first styling |
| Fetch API, localStorage, camera API | REST calls with bearer token; token and user cache; live capture of three facial angles |

# CHAPTER 5 IMPLEMENTATION AND RESULT

## 5.1 Implementation

The application was implemented as a FastAPI back end and a React front end, with a SQLite file selected through the DATABASE_URL entry of the environment file. The steps below follow the user's experience.

- **Step 1 – Registration, login and session.** The front end stores the returned JWT and user record in local storage and attaches the token to every request. On start-up a USER without a profile goes to the wizard; dermatologists and consultants go directly to the clinical portal.
- **Step 2 – Profile wizard.** Skin type and age group, then concerns, then allergies and sensitivities, saved through a create-or-update endpoint and editable later.

- **Step 3 – Lifestyle logging.** A modal form records date, sleep, water, UV, pollution, stress and sunscreen; saving an existing date updates the entry. The overview shows the seven latest logs and averages.
- **Step 4 – Skin texture scan.** Three-angle or single-photo mode, with images captured by camera or uploaded and sent as files or Base64. The server validates the image, runs the analyzer, stores scores, diagnostics and overlays, and returns them; the history lists earlier scans.
- **Step 5 – Assessment.** The server loads the profile, computes statistics from the latest 30 logs, retrieves the latest scan and runs the engine; the overall score, factor breakdown, concerns, risks and lifestyle snapshot are saved as a history snapshot.
- **Step 6 – Routine generation.** Using the latest assessment, profile and scan, the morning, evening, weekly routines and seasonal tips are composed and shown step by step.
- **Step 7 – Recommendation and ingredient safety.** Up to five products ranked by TF-IDF cosine similarity; an ingredient analysis lists safe ingredients and flags allergy matches.
- **Step 8 – Progress, skin cycle, notifications.** Users log score, adherence and notes and view the trend; the cycle tab gives the journey dashboard, cycling plan and turnover tracker; the bell icon opens notifications, replenishment and reminder settings.
- **Step 9 – Dermatologist portal.** The patient panel shows profile, latest assessment and scan, with a prescription form; the prescription is stored and returned to the patient through my-prescription.

## 5.2 Testing

Functional testing exercised the REST API with the FastAPI test client against a copy of the project database, and the front end was inspected in a browser. The assessment cases used a combination skin type, age group 25–34, concerns Acne and Dark Spots, a Fragrance allergy, and a log of 5.5 h sleep, 1,400 ml water, high UV, high stress and no sunscreen.

**Table 5.1 Functional Test Cases and Observed Results**

| No. | Test case | Observed result (all passed) |
|---|---|---|
| 1 | Health check, GET /health | {"status": "healthy"} |
| 2 | Database connectivity, GET /database-test | Database connected |
| 3 | Register a new user | 201 with access token |
| 4 | Login with valid credentials | 200 with JWT |
| 5 | Login with wrong password | 401 Unauthorized |
| 6 | Access /api/auth/me without token | 401 Unauthorized |
| 7 | Create skin profile | 200, profile saved |
| 8 | Save a lifestyle log | 200, log saved |
| 9 | Run skin assessment | 201; overall score 49.0; 5 risk factors |
| 10 | Generate routine | 201; 4 morning and 3 evening steps |
| 11 | Product recommendations | 200; 5 products returned |
| 12 | Ingredient analysis (Fragrance allergy) | No flagged ingredients for tested product; marked safe |
| 13 | Log progress entry, read analytics | 201; trend neutral for a single entry |
| 14 | Retrieve notifications | 200; 3 unread notifications synchronized |
| 15 | Analyze uploaded facial image | 200; scores and overlays stored |

**Manual verification of test 9.** With no scan, Skin Condition = 90 − (2 × 6) = 78, contributing 27.30. Lifestyle = 80 − 20 − 15 = 45, contributing 9.00. Sleep of 5.5 h gives 55, contributing 8.25. Sunscreen compliance of 0% contributes 0. Water of 1,400 ml gives 45, contributing 4.50. The sum is 49.05, reported as 49.0 (“needs dedicated care”), matching the system. A milestone script also runs register → profile → assessment → routine in sequence, Pydantic enforces validation (password ≥ 6 characters, sleep 0–24 h, non-negative water), and texture endpoints reject requests without a valid image.

## 5.3 Results

The screens were captured from the running application using a demonstration account: combination skin, age 25–34, concerns Acne and Dark Spots, an Alcohol sensitivity and a Fragrance allergy, with three days of lifestyle logs, a facial scan from the supplied sample image and several progress entries.

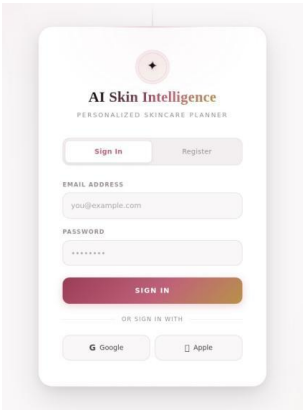



Fig. 5.1 Login Screen

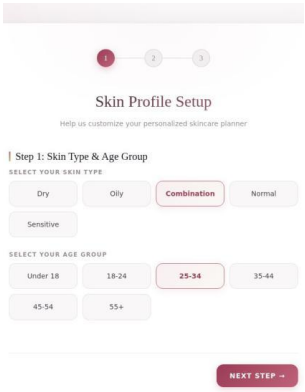



Fig. 5.2 Skin Profile Setup Wizard

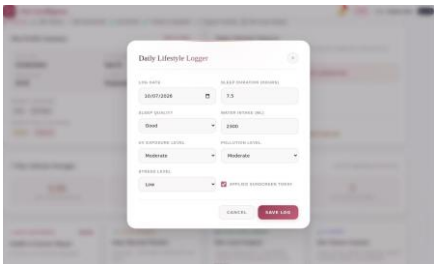



Fig. 5.3 Daily Lifestyle Logger

Fig. 5.1 shows the combined sign-in/registration screen with simulated Google and Apple buttons; a new USER is directed to the wizard (Fig. 5.2). The overview tab summarizes the profile and shows seven-day averages of 6.8 hours of sleep, 2.0 litres of water and 67% sunscreen compliance across three logged days, with shortcuts to assessment, routine, skin cycle and scanner. The logger (Fig. 5.3) is validated by the server, and a second save on the same date updates the entry.

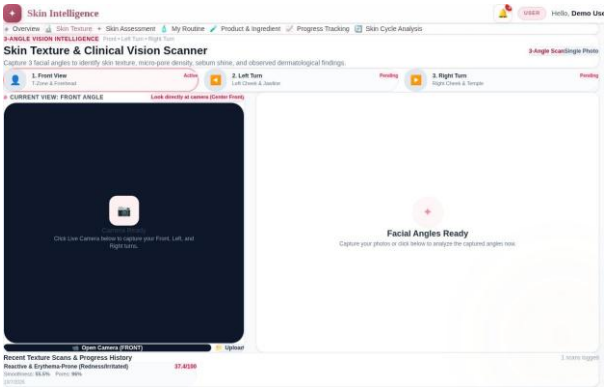



Fig. 5.4 Skin Texture & Clinical Vision Scanner

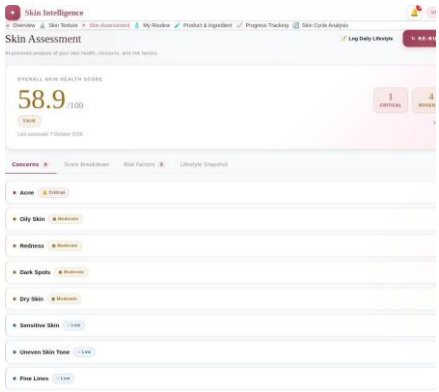



Fig. 5.5 Skin Assessment Panel

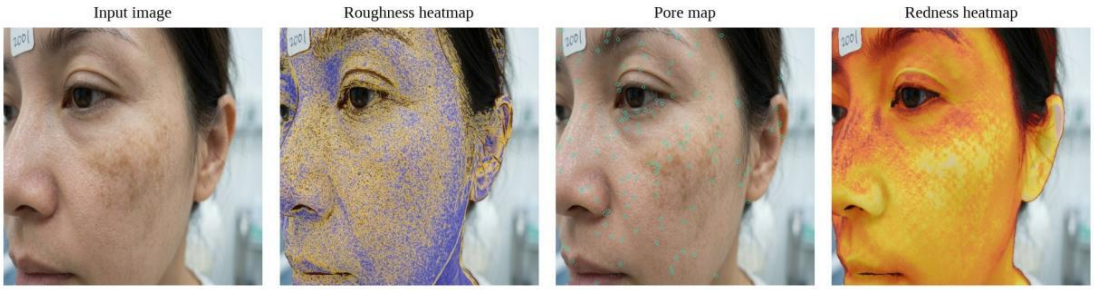



**Fig. 5.6 Image Analysis Outputs (input, roughness heatmap, pore map, redness heatmap)**

The scanner (Fig. 5.4) offers three-angle capture steps, a camera panel and recent scan history. The analyzer outputs for the sample image (Fig. 5.6) are listed in Table 5.2.

**Table 5.2 Texture Scores for the Sample Image**

| Metric | Score (0–100) |
|---|---|
| Smoothness / Roughness | 55.5 / 44.5 |
| Pore visibility / Pore density | 96.0 / 95.0 |
| Oiliness / shine | 55.9 |
| Redness / erythema | 91.9 |

| Metric | Score (0–100) |
|---|---|
| Fine lines | 21.7 |
| Overall texture score | 37.4 |
| Texture type | Reactive & Erythema-Prone (Redness / Irritated) |

The assessment (Fig. 5.5) gave an overall score of **58.9 / 100 (fair)** with 1 critical, 4 moderate and 3 low concerns and 5 risk factors. Acne scored highest at 8/10, followed by Oily Skin and Redness at 6, Dark Spots at 5 and Dry Skin at 4; each concern expands to show its modifiers. The high Oily Skin and Redness scores come from the scan's pore, oiliness and redness measurements, showing the integration of the vision module and the assessment engine. Fig. 5.7 and Table 5.3 give each factor's contribution.

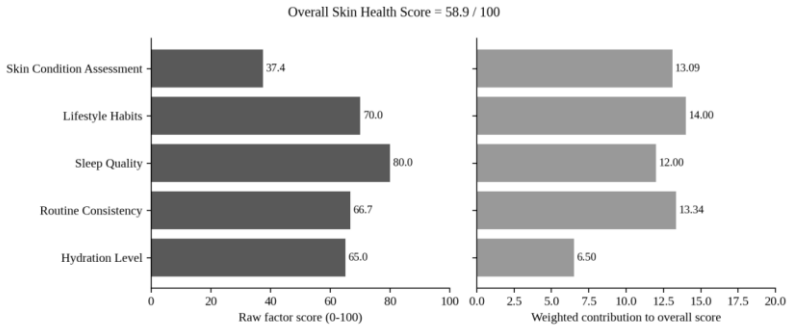



**Fig. 5.7 Score Breakdown of the Skin Health Score**

**Table 5.3 Five-Factor Breakdown for the Demonstration Account**

| Factor | Weight | Raw score | Weighted contribution |
|---|---|---|---|
| Skin Condition Assessment | 35% | 37.4 | 13.09 |
| Lifestyle Habits | 20% | 70.0 | 14.00 |
| Sleep Quality | 15% | 80.0 | 12.00 |
| Routine Consistency | 20% | 66.7 | 13.34 |
| Hydration Level | 10% | 65.0 | 6.50 |
| **Overall Skin Health Score** | 100% | – | **58.9** |

The risk factors were Compromised Surface Texture and Barrier (high), Elevated Facial Erythema or Reactivity (high), Sebaceous Congestion and Pore Clustering (moderate), Suboptimal Skin Recovery (low) and Mild Dehydration (low); the first three arise from the facial scan and the last two from lifestyle averages.

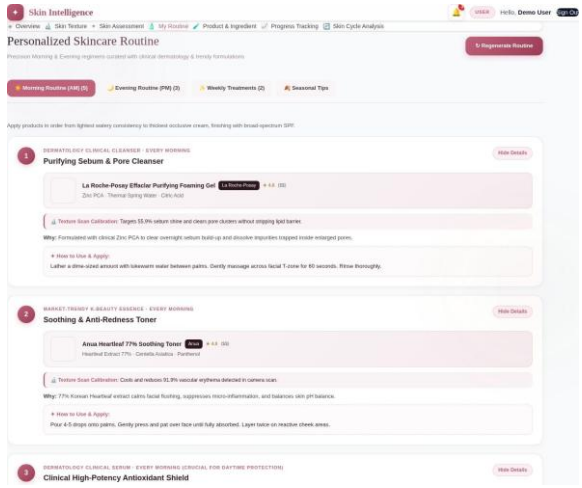



**Fig. 5.8 Personalized Skincare Routine**

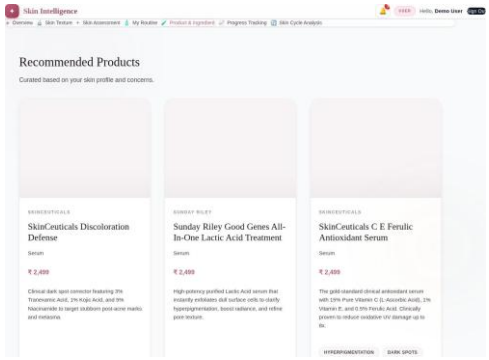



**Fig. 5.9 Product Recommendations**

Each routine step (Fig. 5.8) shows the category, product and brand, a clinical or trendy label, the targeted concern, the reason and the application instructions, with separate morning, evening, weekly and seasonal views. The five recommended products (Fig. 5.9) were SkinCeuticals Discoloration Defense, Sunday Riley Good Genes All-In-One Lactic Acid Treatment, SkinCeuticals C E Ferulic Antioxidant Serum, EltaMD UV

Clear Broad-Spectrum SPF 46 and La Roche-Posay Effaclar Purifying Foaming Gel; the first three match Dark Spots and the last two Acne, so the ranking follows the profile concerns.

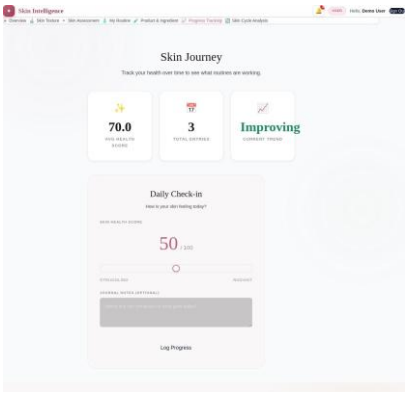



**Fig. 5.10 Progress Tracking**

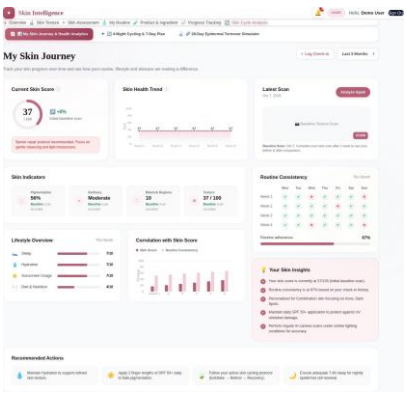



**Fig. 5.11 Skin Cycle Analysis**

After three progress entries of 66, 70 and 74 (Fig. 5.10), the analytics endpoint returned an average of 70.0, three entries and an "Improving" trend, as the last score exceeds the first. The Skin Cycle Analysis tab (Fig. 5.11) displays the scan score, health trend, routine consistency calendar, lifestyle overview, lifestyle–score correlation and insights, plus the four-night protocol and 28-day turnover tracker.

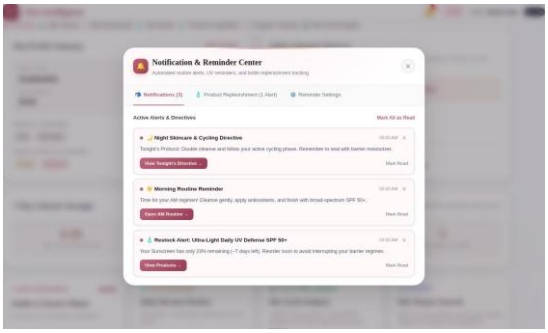



**Fig. 5.12 Notification and Reminder Centre**

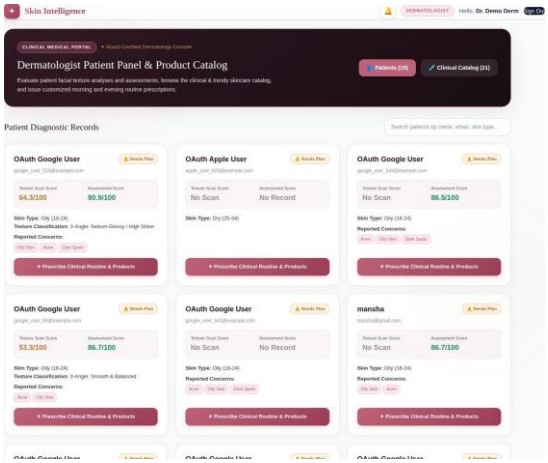



**Fig. 5.13 Dermatologist Clinical Portal**

The notification centre (Fig. 5.12) shows a night-time cycling directive, a morning reminder and a restock alert for a sunscreen with 23% remaining; each offers an action button to the relevant tab. The dermatologist portal (Fig. 5.13) lists each patient's skin type, latest assessment and texture scores, texture type and concerns, with a button to prescribe a clinical routine and product plan.

## 5.4 Database Contents

The delivered database holds the demonstration and test accounts, the 21-product catalogue, assessments and routines from repeated runs, eight texture scans and one clinical prescription: users 16, skin_profiles 14, lifestyle_logs 5, skin_assessments 20, skincare_routines 19, skin_texture_analyses 8, products 21, progress_logs 5, notifications 6, notification_settings 2, product_replenishments 6 and dermatologist_prescriptions 1. Of the 16 accounts, 14 are USER, one is a dermatologist and one a skincare consultant. The catalogue comprises 9 serums, 3 cleansers, 3 moisturizers, 3 sunscreens, 2 exfoliants and 1 toner (11 Dermatology Clinical, 10 Market Trendy).

## 5.5 Offline Machine-Learning Experiments

A generator script creates a synthetic dataset of 10,000 samples (skin type, age group, sleep, stress, water, UV, sunscreen rate) with four concern labels and a health score from probabilistic rules. Label prevalences are 33.9% acne, 21.8% dry skin, 23.2% wrinkles and 24.5% dark spots; the score has mean 84.4 and range 48–100. Two scikit-learn pipelines apply StandardScaler to numeric and OneHotEncoder to categorical features: an XGBoost regressor (200 estimators, learning rate 0.05, depth 6) for the score and a multi-output

XGBoost classifier with the same settings for the four labels. Data are split 80:20 (8,000 train, 2,000 test, seed 42) and the pipelines are saved with joblib. Results are in Table 5.4 and Fig. 5.14.

**Table 5.4 Offline XGBoost Experiment Results on the Synthetic Dataset**

| Model / label | Metric | Value |
|---|---|---|
| Health score regressor | MSE / RMSE / MAE | 43.86 / 6.62 / 5.40 |
| Health score regressor | $R^2$ | 0.492 |
| Concern classifier (all four labels) | Exact-match accuracy | 0.426 |
| Acne | Accuracy / F1-score | 0.724 / 0.541 |
| Dry skin | Accuracy / F1-score | 0.849 / 0.623 |
| Wrinkles | Accuracy / F1-score | 0.842 / 0.614 |
| Dark spots | Accuracy / F1-score | 0.796 / 0.478 |

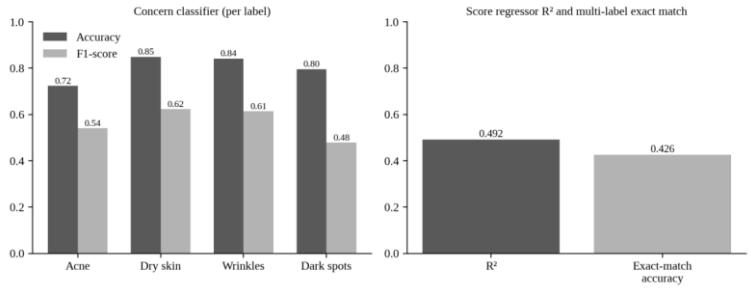



**Fig. 5.14 Offline Machine-Learning Experiment Results**

The moderate values are expected because labels were generated by adding random draws to the probabilities, so much of the variation cannot be predicted from the attributes. The experiment is therefore an offline exploration on synthetic data. In the delivered application the live assessment is performed by the explainable rule-based engine, and the only trained model used at run time is the TF-IDF vectorizer of the product recommender.

# CHAPTER 6 CONCLUSION AND FUTURE ENHANCEMENT

## 6.1 Conclusion

An AI Skin Intelligence and Personalized Skincare Planner has been developed as a full-stack web application using FastAPI, SQLAlchemy, SQLite and React. It accepts a skin profile and daily lifestyle logs, measures facial skin texture from single or three-angle images, and converts these inputs into an explainable skin health score, ranked concerns and risk factors, and a personalized morning, evening and weekly routine. A TF-IDF recommender and allergy-aware ingredient check provide product guidance, while progress tracking, skin-cycling and turnover views and a notification centre support continued use. A dermatologist portal allows a clinician to review patient data and issue a prescribed regimen.

Functional testing confirmed correct authentication, successful execution of the complete user workflow, and agreement between the implemented scoring model and a manual calculation. The captured screens showed that the profile, lifestyle, vision, assessment, routine, recommendation, progress, notification and clinical modules work together. Live decision logic stays transparent, every score can be traced to its inputs, and the offline experiments provide a starting point for future data-driven work.

## 6.2 Limitations

- Image analysis uses colour and gradient heuristics and is sensitive to lighting, camera quality and framing, so scores are indicative rather than clinical.
- The catalogue is limited to 21 products and the routine generator uses a fixed product set.
- The XGBoost models were trained on synthetic data and are not connected to the live assessment; the social sign-in is simulated.
- Dermatologist endpoints identify the caller by token, but restriction of the clinical portal to clinical roles is currently applied only in the front end.
- The system is a planning aid and does not provide a medical diagnosis.

### 6.3 Future Enhancement

- **Server-side role enforcement:** role checks on all dermatologist and administrative endpoints, trusted-origin CORS, and a secret key required from the environment with no built-in default.
- **Real social sign-in:** replace the simulation with standard OAuth 2.0 providers.
- **Learned models:** train on real, consented data, validate against clinician annotations and integrate with the rule-based engine.
- **Improved vision analysis:** trained image models, calibrated lighting normalization, automatic face detection and capture guidance.
- **Larger product knowledge base:** expand the catalogue, link routine steps to entries, and consider price, availability and ratings.
- **Deployment and notifications:** production database, cloud hosting, and push, e-mail or messaging reminders.
- **Automated testing:** a complete unit and integration test suite with continuous integration.

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